

Room Calling System

Software Requirements Specification (SRS) & Technical Design Document
Version 1.0

1. Executive Summary

The Room Calling System is a distributed real-time alert and monitoring platform designed for hospitals, care centers, wards, and supervisory stations. The architecture follows a Single Source of Truth (SSOT) model where a Server device manages all room data while Client and Master devices synchronize through WebSockets.

2. Objectives

- Real-time room call monitoring
- Centralized data management
- Low network bandwidth usage
- Multi-station supervision
- Emergency escalation handling
- High availability and automatic recovery

3. System Roles

Server: Database owner, web dashboard host, WebSocket hub.

Client: Visual/audio display synchronized with a single server.

Master: Supervisory display connected to multiple servers simultaneously.

4. High Level Architecture

Room Trigger -> Server(SQLite + WebSocket + Web Dashboard) -> Clients & Masters

All updates originate from the Server and are broadcast to connected devices.

5. Startup Flow

Application launch checks saved AppType.

If role exists, load role module directly.

If role not selected, display Server / Client / Master selection screen.

Triple-tap gesture resets role configuration.

6. Server Module Design

Responsibilities:

- Maintain SQLite database.
- Host NanoHTTPD web dashboard on port 8080.
- Host WebSocket server on port 8090.
- Broadcast state changes.

Initialization:

1. Start SQLite.
2. Start NanoHTTPD.
3. Start WebSocket Server.
4. Detect local IP.
5. Display Help URL.

Business Rules:

- Room creation only via browser dashboard.
- Any database update triggers REFRESH_ALL broadcast.
- Server acts as authoritative source.

7. Web Dashboard

Features:

- Room Add
- Room Edit
- Room Delete
- Room Status View

URL:

`http://<server-ip>:8080/help`

Security Recommendations:

- Local network access only.
- Optional dashboard password.

8. Client Module Design

Connection Lifecycle:

1. Request server IP.
2. Connect to `ws://server-ip:8090`.
3. Auto reconnect every 5 seconds.
4. Send GETDETAIL after successful connection.

UI:

- Grid layout.
- Real-time synchronization.

Audio Engine:

- Detect highest active priority.
- Play one sound only.
- Stop playback when all rooms return idle.

9. Master Module Design

Multi-Server Features:

- Multiple WebSocket connections.
- Individual station cache.
- Automatic station rotation.

Rotation Engine:

- 10-second station cycle.
- Manual selection resets timer.

Emergency Handling:

- Status 4 interrupts all workflows.
- Full-screen emergency overlay.
- Auto dismiss after resolution or timeout.

10. Communication Protocol

Heartbeat:

o

Room Status Update:

s[RoomID],[Status]

Example:

s101,1

Full Synchronization:

REFRESH_ALL

Recommended JSON Payload:

```
{
  "action": "REFRESH_ALL",
  "station": "Ward A",
```

```
"data":[...]  
}
```

11. Status Definitions

0 = Idle
1 = Call
2 = Assistance
3 = Care Required
4 = Emergency / Code Blue

12. Database Design

Table: Rooms

RoomID (PK)
RoomName
StationName
Status
LastUpdated

Indexes:

- RoomID
- StationName
- Status

13. Reconnection Strategy

Client:

- Retry every 5 seconds.

Master:

- Reconnect each station independently.

Server:

- Preserve active connections list.
- Remove dead sockets automatically.

14. Error Handling

Network Failure:

- Show disconnected indicator.
- Retry automatically.

Server Offline:

- Maintain cached data.
- Display last sync timestamp.

Database Error:

- Log exception.
- Notify administrator.

15. Performance Requirements

Maximum Room Refresh Latency: < 1 second

Heartbeat Interval: 15-30 seconds

Client Reconnect Interval: 5 seconds

Master Rotation Interval: 10 seconds

16. Security Requirements

- Local network communication.
- Input validation.
- Web dashboard access control.
- Audit logging for administrative actions.

17. Auto Start Requirements

Android BOOT_COMPLETED receiver launches server service automatically after device reboot.

18. Testing Checklist

- ✓ Server startup
- ✓ Client connection
- ✓ Auto reconnect
- ✓ Room status synchronization
- ✓ Emergency popup
- ✓ Station rotation
- ✓ Audio priority
- ✓ Dashboard CRUD operations
- ✓ Boot auto-start
- ✓ Network recovery

19. Future Enhancements

- User authentication
- Cloud synchronization

- Push notifications
- Reporting dashboard
- Analytics
- Device health monitoring